ScolioLDM: Towards Accurate Postoperative Radiograph Synthesis for Scoliosis with Morphology-Guided Latent Diffusion Model

***Abstract*—Accurate prediction of postoperative morphology is critical for surgical planning in adolescent idiopathic scoliosis (AIS). Current planning workflows predominantly rely on quantitative indices, which provide only numerical references and offer limited intuitive visualization of postoperative spinal alignment and implant configuration, thereby constraining precise decision-making. Image-level postoperative outcome prediction from preoperative radiographs is a promising solution, yet remains challenging due to non-rigid spinal deformation and the complex geometry of instrumentation. To address these challenges, we propose ScolioLDM, a spinal morphology–guided conditional latent diffusion framework that synthesizes postoperative radiographs directly from preoperative radiographs, enabling a new paradigm for image-level prediction of AIS surgical outcomes. To enforce anatomical and geometric consistency during diffusion, we design a Morphology-Guided Control Module (MGCM) that injects deformity-aware morphology features via morphology-conditioned control. In addition, we construct the first dataset of paired pre- and post-operative radiographs for AIS. Using only a single pre-operative image as input, ScolioLDM can generate realistic post-operative radiographs. Experimental results demonstrate consistent improvements over many mainstream and advanced generative approaches in image fidelity, providing an intuitive and practical tool for AIS surgical planning.**

***Index Terms*—Adolescent idiopathic scoliosis, latent diffusion model, morphology guidance, postoperative radiograph synthesis, surgical planning.**

I. Introduction

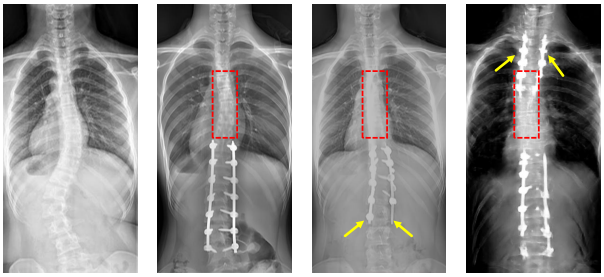
ADOLESCENT idiopathic scoliosis (AIS) is one of the most common types of spinal deformity, characterised by insidious onset and difficulty in early detection [1]. This condition can rapidly progress to moderate or severe deformity, leading to cosmetic deformity, limited mobility, pain, and impaired cardiopulmonary function [2]. For cases with progressive disease, surgical correction remains the most effective method for restoring spinal balance and improving quality of life [3]. However, the complex anatomy of the spine and the strong biomechanical coupling between segments require surgeons to consider multisegmental three-dimensional (3D) deformities, spinal rotation, and implant configuration collectively during preoperative planning [4]. This process is time-consuming, technically demanding, and highly operator-dependent, leading to significant variability in correction quality and postoperative alignment [5]. Consequently, there is an urgent need for methods capable of objectively and accurately predicting postoperative outcomes to support more reliable surgical planning.

Accurate prediction of postoperative spinal morphology and alignment is a key component of AIS surgical planning [6]. In current clinical practice, surgeons primarily rely on quantitative measurements, such as the Cobb angle [7], and experience-based judgements to estimate postoperative alignment; these methods are inherently subjective and lack uniform quantitative standards [5]. Early attempts at automation relied primarily on finite element simulations or biomechanical models [8]. Although these methods offer strong biophysical interpretability, they typically require complex modelling processes, numerous parameters, and high computational costs, and are difficult to adapt flexibly to patient-specific anatomy and diverse surgical strategies [9], which limits their widespread clinical application.

Recently, deep learning has made significant progress in spinal image analysis, particularly in anatomical segmentation and key point detection [10]. However, most existing methods focus on the regression of specific scalar metrics, such as the automatic measurement of the Cobb angle [11], or the classification of disease progression. Although these models are useful for basic clinical assessment and risk stratification, their ability to support preoperative planning, such as fusion level selection, correction strategy design, and screw placement, remains limited. [28] They fail to effectively capture global spinal deformation, thoracic contour changes, and postoperative implant configurations. Moreover, prediction methods based on only a few scalar metrics are often sensitive to distribution variations and lack robustness and interpretability. Therefore, image-level prediction of postoperative outcomes is essential for more intuitive and reliable surgical planning.

From the perspective of image-level prediction and generation, AIS surgery introduces a significant amount of non-rigid deformation, involving complex couplings between vertebral rotation, coronal/sagittal alignment, and local structural changes caused by metal implants. Mainstream conditional generative models, such as pix2pix [13] under generative adversarial network (GAN) architectures and DDIB [12] under diffusion model architectures, are primarily applied to cross-modal medical imaging tasks, e.g., CT-MR conversion, where anatomical morphology remains relatively constant. When directly applied to in-modality transformations between preoperative and postoperative spinal radiographs, these models often fail to capture the complex changes in bone and overall spinal shape, resulting in blurred structural boundaries, increased artefacts, and poor anatomical consistency (see Fig. 1) [14]. These limitations severely restrict their applicability to image-guided preoperative decision-making.

To address the limitations of existing methods, this paper proposes ScolioLDM, a morphology-guided latent diffusion model (LDM) for postoperative radiograph synthesis in AIS. We introduce the curve map as a morphological prior and



(a) preoperative radiograph (b) postoperative radiograph (GT) (c) DDIB (d) pix2pix

Fig. 1. Illustration of postoperative radiograph synthesis directly using mainstream conditional generative models. Both the diffusion-based model (e.g., DDIB [12]) and GAN-based model (e.g., pix2pix [13]) fail to reconstruct the spinal anatomical structure (see red square) and implant configuration (see yellow arrow), exhibiting severe artifacts or anatomical distortions when compared to the real postoperative radiograph (subfigure (b)).

design a Morphology-Guided Control Module (MGCM) to incorporate structural information into the diffusion process. By jointly constraining spinal anatomy and curve morphology in the latent space, ScolioLDM captures the complex spinal transformation from preoperative to postoperative states while preserving implant structures. As a result, the proposed framework generates anatomically consistent and visually realistic postoperative radiographs with accurate spinal alignment and implant placement. To the best of our knowledge, ScolioLDM is the first generative framework for image-level scoliosis surgical outcome prediction through postoperative radiograph synthesis.

II. Method

A. Diffusion Preliminaries

We adopt a LDM to synthesize postoperative radiographs. Let X_{post} denote the ground truth (GT) postoperative radiograph. Following the encoder of the latent autoencoder, X_{post} is mapped to a latent variable $x_0 = \mathcal{E}(X_{\text{post}})$, and the final radiograph is obtained by decoding $\hat{X}_{\text{post}} = \mathcal{D}(\hat{x}_0)$.

1) Forward process: A variance schedule $\{\beta_t\}_{t=1}^{T}$ defines $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. The forward diffusion progressively perturbs x_0 into Gaussian noise:

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}\, x_0, (1 - \bar{\alpha}_t)\mathbf{I}\right). \tag{1}$$

Equivalently, x_t can be sampled using the reparameterization $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

2) Reverse process: The denoising network ϵ_θ learns to predict the injected noise conditioned on timestep t and condition c, defining the reverse transitions $p_\theta(x_{t-1} \mid x_t, c)$. We use the standard ϵ-parameterization, where $\epsilon_\theta = \epsilon_\theta(x_t, t, c)$.

3) Training objective: The model is trained with the noise-prediction loss:

$$\mathcal{L} = \mathbb{E}_{t,\epsilon,x_0}\left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|_2^2\right]. \tag{2}$$

4) Sampling: Starting from $x_T \sim \mathcal{N}(0, \mathbf{I})$, we iteratively denoise to obtain $\hat{x}_0$. A common DDPM sampling update is:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_\theta(x_t, t, c)\right) + \sigma_t z, \tag{3}$$

where σ_t is determined by the chosen variance schedule, and $z \sim \mathcal{N}(0, \mathbf{I})$.

In our setting, the condition is defined as $c = \{X_{\text{pre}}, \hat{S}_{\text{post}}\}$, where X_{pre} is the preoperative radiograph and $\hat{S}_{\text{post}}$ is the predicted postoperative spine curve map.

B. Postoperative Curve Map Prediction

To provide explicit morphological guidance for postoperative radiograph synthesis, we first predict the postoperative curve map from the preoperative structural condition. Given the preoperative radiograph X_{pre} and its corresponding curve map S_{pre}, a ControlNet-based prediction network is employed to generate the expected postoperative structural map:

$$\widehat{S}_{\text{post}} = F_{\text{CN}}\left(X_{\text{pre}}, S_{\text{pre}}\right), \tag{4}$$

where $F_{\text{CN}}(\cdot)$ denotes the ControlNet-based curve-map prediction network [20]. The preoperative curve map provides explicit geometric conditioning on spinal curvature and vertebral alignment, enabling the network to learn the structural transformation induced by surgical correction. The predicted postoperative curve map $\widehat{S}_{\text{post}}$ is then used as morphology guidance for subsequent postoperative radiograph synthesis in ScolioLDM.

C. Postoperative Radiograph Synthesis

After obtaining $\widehat{S}_{\text{post}}$, ScolioLDM uses X_{pre} and $\widehat{S}_{\text{post}}$ as inputs. The structural map provides dual guidance: it is processed through the feature extractor, MGCM, and zero convolution as a morphological condition, and is also encoded and concatenated with X_{pre} features as an image condition. X_{pre} supplies patient-specific appearance information. With these image and morphological conditions, the denoising U-Net generates the postoperative radiograph at time step t.

1) MGCM-Based Morphology Injection Branch: Given the predicted postoperative black-background white-line image

$$M = \widehat{S}_{\text{post}}, \tag{5}$$

we first extract its structural representation using the feature extractor $E_f(\cdot)$:

$$x = E_f(M), \qquad x \in \mathbb{R}^{B \times C \times H \times W}, \tag{6}$$

where B, C, H and W denote the batch size, number of feature channels, spatial height, and spatial width of the extracted structural feature map, respectively.

To further encode postoperative spinal morphology and spatial deformation patterns, the extracted feature x is fed into the MGCM, whose detailed architecture is illustrated in Fig. 3. MGCM contains a deform refinement branch, a spatial gating branch, a deformation field prediction branch, and a differentiable warping operation.

First, the deform refinement branch enhances the local structural representation of the input feature:

$$F_r = \mathcal{R}(x), \qquad F_r \in \mathbb{R}^{B \times C \times H \times W}, \tag{7}$$

where $\mathcal{R}(\cdot)$ denotes the deform refinement mapping composed of depthwise convolution, pointwise convolution, group normalisation, and SiLU activations, as shown in Fig. 3. This

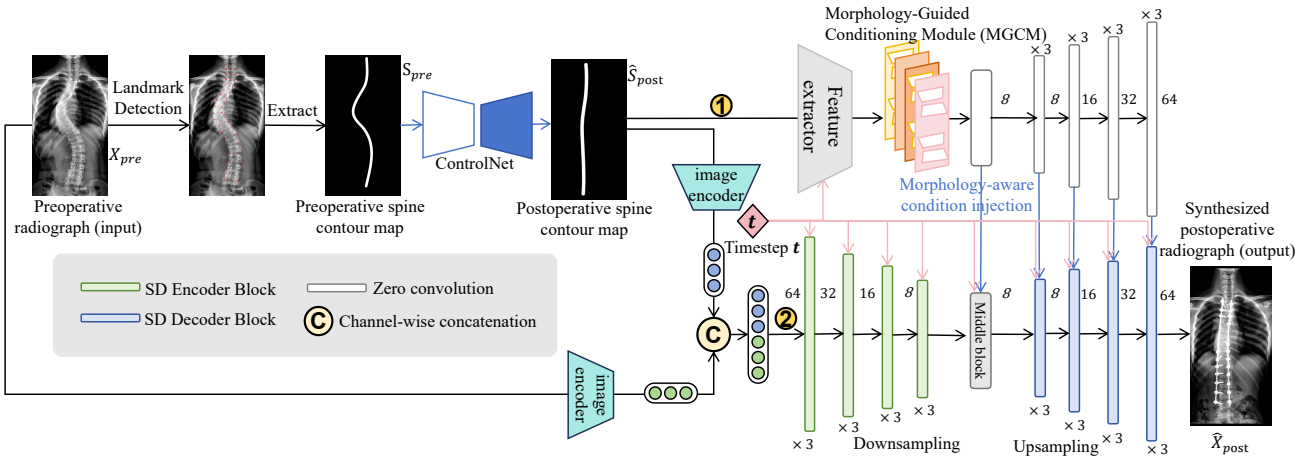


Fig. 2. Overview of the proposed ScolioLDM. The left part presents the overall framework for postoperative radiograph synthesis, while the right part illustrates the detailed architecture of the core denoising U-Net integrated with the MGCM. ControlNet [20] is used to introduce additional structural conditions into the diffusion model for controllable postoperative radiograph synthesis [29] .

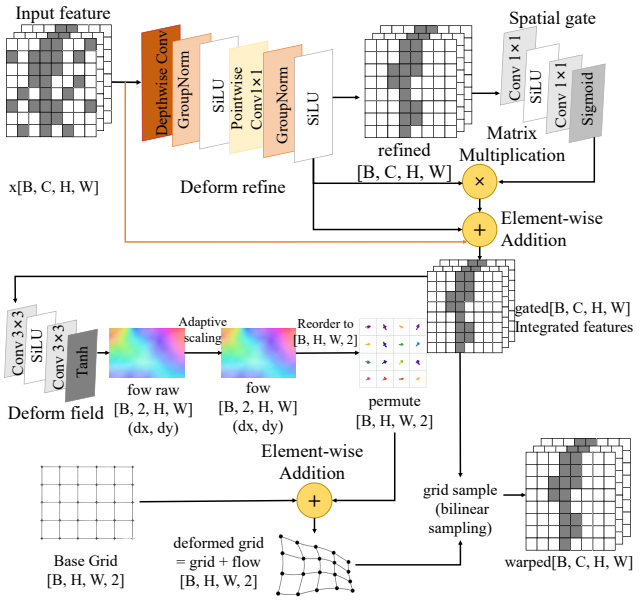


Fig. 3. Detailed illustration of the Morphology-Guided Control Module (MGCM) embedded in the proposed ScolioLDM framework.

operation improves the representation of local spinal curvature, vertebral orientation, and fine-grained morphological details.

Based on the refined feature F_r, the spatial gating branch predicts a spatial attention map A:

$$A = \mathcal{G}(F_r), \qquad A \in \mathbb{R}^{B\times 1\times H\times W}, \tag{8}$$

where $\mathcal{G}(\cdot)$ denotes the spatial gate illustrated in Fig. 3. The gate selectively emphasises structurally informative regions while suppressing irrelevant background responses.

The refined feature is then modulated by the spatial gate and fused with the original structural feature through a residual connection:

$$F_g = F_r \odot A + x, \qquad F_g \in \mathbb{R}^{B\times C\times H\times W}, \tag{9}$$

where $\odot$ denotes element-wise multiplication. In this way, F_g preserves the basic morphological information in x while incorporating spatially enhanced structural cues.

Subsequently, the deformation field prediction branch estimates a two-channel offset field from the gate-enhanced feature:

$$\Delta P = \alpha\mathcal{D}(F_g), \qquad \Delta P \in \mathbb{R}^{B\times 2\times H\times W}, \tag{10}$$

where $\mathcal{D}(\cdot)$ denotes the deformation field predictor shown in Fig. 3, and α is a scaling factor used to constrain the magnitude of the predicted offsets. In our implementation, α is set to 0.1. The two channels of ΔP represent horizontal and vertical spatial displacements, respectively.

To construct the sampling coordinates, the predicted deformation field is rearranged and added to the standard sampling grid:

$$P_d = P_0 + \Pi(\Delta P), \qquad P_d \in \mathbb{R}^{B\times H\times W\times 2}, \tag{11}$$

where $P_0 \in \mathbb{R}^{B\times H\times W\times 2}$ denotes the base sampling grid, and $\Pi(\cdot)$ denotes the permutation operation that converts the offset field to the grid sampling format.

Finally, the gate-enhanced feature is spatially warped according to the deformed sampling grid using bilinear interpolation:

$$F_w = \mathrm{GridSample}(F_g, P_d), \qquad F_w \in \mathbb{R}^{B\times C\times H\times W}. \tag{12}$$

The resulting feature F_w serves as the morphology-guided representation, which incorporates both postoperative structural saliency and deformation-aware spatial information.

2) Image Feature Fusion Branch: The predicted postoperative structural map $\widehat{S}_{\mathrm{post}}$ is further encoded to obtain structural image features:

$$z_s = E_s\left(\widehat{S}_{\mathrm{post}}\right), \tag{13}$$

while the preoperative radiograph X_{pre} is encoded to extract patient-specific appearance features:

$$z_{pre} = E_p\left(X_{\mathrm{pre}}\right). \tag{14}$$

Here, z_s provides the postoperative spatial prior, and z_{pre} preserves the anatomical appearance and imaging characteristics of the patient. These features are concatenated along the channel dimension:

$$z_c = \mathrm{Concat}\left(z_s, z_{\mathrm{pre}}\right). \tag{15}$$

The fused feature z_c serves as the image-conditional input to the denoising U-Net, enabling joint guidance from preoperative appearance and postoperative structure.

3) Postoperative Radiograph Output: The ScolioLDM denoising U-Net fuses three inputs: noisy diffusion latent x_t, image conditional concatenated features z_c, and multi-scale structural control features $\mathbf{O} = \{O_i\}|_{i=1}^{K}$ from MGCM with zero convolution. The denoising formulation reads:

$$\widehat{x}_0 = U_\theta\left(x_t, t, z_c, \mathbf{O}\right), \tag{16}$$

where t is the diffusion timestep, z_c encodes pre- and postoperative image priors, and $\mathbf{O}$ imposes multi-scale shape constraints. The decoder $D(\cdot)$ projects the denoised latent $\widehat{x}_0$ into image space, yielding the predicted postoperative radiograph $\widehat{X}_{\mathrm{post}} = D(\widehat{x}_0)$.

$\widehat{X}_{\mathrm{post}}$ is bounded by preoperative radiograph X_{pre} for personalized anatomy retention and guided by predicted structural sketch $\widehat{S}_{\mathrm{post}}$ to match target spinal correction geometry.

III. Experiments and Results

A. Data Collection and Preprocessing

From June 2023 to August 2025, 612 eligible patients with AIS who underwent posterior corrective fusion surgery were retrospectively enrolled from Queen Mary Hospital. This study was approved by the Institutional Review Board (IRB No. UW 25-300, July 8, 2025), and informed consent was obtained from all patients and/or their guardians. Patients aged 10–18 years with clinically and radiographically confirmed AIS, complete surgical records, preoperative and postoperative spinal posteroanterior (PA) radiographs, and surgical planning data were included. Patients with other spinal deformities, previous spinal surgery, or poor-quality radiographs were excluded.

Raw DICOM images were converted to 8-bit grayscale, resized to 240×480 pixels, and preprocessed using adaptive contrast enhancement and Gaussian denoising ($\sigma = 1.0$, 3×3 kernel). Preoperative ROI masks were independently annotated by two senior spine surgeons, achieving strong inter-observer agreement ($\kappa > 0.85$). Vertebral landmark annotations from preoperative and postoperative PA radiographs were used to generate curve maps representing spinal morphology before and after correction.

The dataset was split into training, validation, and test sets at a ratio of 8:1:1. GT annotations, including implant masks, fusion-level labels, geometric parameters, and postoperative curve maps, were provided by senior spine surgeons and imaging analysts for model training and evaluation.

B. Evaluation Metrics

To evaluate the quality of generated postoperative radiographs for AIS patients, we use five metrics from four aspects: pixel-wise reconstruction accuracy, signal fidelity, structural and edge preservation, and gray-scale correlation. Let Y_i and $\widehat{Y}_i$ denote the GT and generated postoperative radiographs of the ith test sample, respectively. Each image contains M pixels, and p denotes the pixel location.

1) Pixel-wise Reconstruction Error and Signal Fidelity: MAE is used to measure the average pixel-wise reconstruction error:

$$\mathrm{MAE}_i = \frac{1}{M}\sum_{p=1}^{M}\left|\widehat{Y}_i(p) - Y_i(p)\right|. \tag{17}$$

PSNR evaluates the image fidelity of predicted postoperative radiographs [30].

2) Structural Similarity and Edge Preservation: SSIM evaluates structural similarity between the synthesized and GT radiographs [31].

GradMAE is used to evaluate edge preservation in spinal contours and fixation structures:

$$\begin{aligned}\mathrm{GradMAE}_i = \frac{1}{2M}\sum_{p=1}^{M}\Big(&\left|\nabla_x\widehat{Y}_i(p) - \nabla_x Y_i(p)\right| \\ &+ \left|\nabla_y\widehat{Y}_i(p) - \nabla_y Y_i(p)\right|\Big),\end{aligned} \tag{18}$$

where ∇_x and ∇_y denote the horizontal and vertical gradient operators, respectively.

3) Overall Gray-Scale Correlation: NCC is used to evaluate the gray-scale correlation between the generated and GT images:

$$\mathrm{NCC}_i = \frac{\sum_{p=1}^{M}\left(\widehat{Y}_i(p) - \overline{\widehat{Y}}_i\right)\left(Y_i(p) - \overline{Y}_i\right)}{\sqrt{\sum_{p=1}^{M}\left(\widehat{Y}_i(p) - \overline{\widehat{Y}}_i\right)^2}\sqrt{\sum_{p=1}^{M}\left(Y_i(p) - \overline{Y}_i\right)^2} + \varepsilon}. \tag{19}$$

Lower MAE and GradMAE values indicate smaller reconstruction errors and better edge preservation, whereas higher PSNR, SSIM, and NCC values indicate better signal fidelity, structural consistency, and gray-scale correlation.

C. Details of Experimental Settings

All experiments were conducted on a single NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. The models were implemented using PyTorch and PyTorch Lightning. Due to GPU memory constraints, the batch size was set to 8. All models were trained using the Adam optimizer with a fixed learning rate throughout training. Model performance was monitored on the validation set during training. Training was terminated when the validation performance converged and no further improvement was observed, while the test set was used only for final evaluation.

D. Controlled Experiment

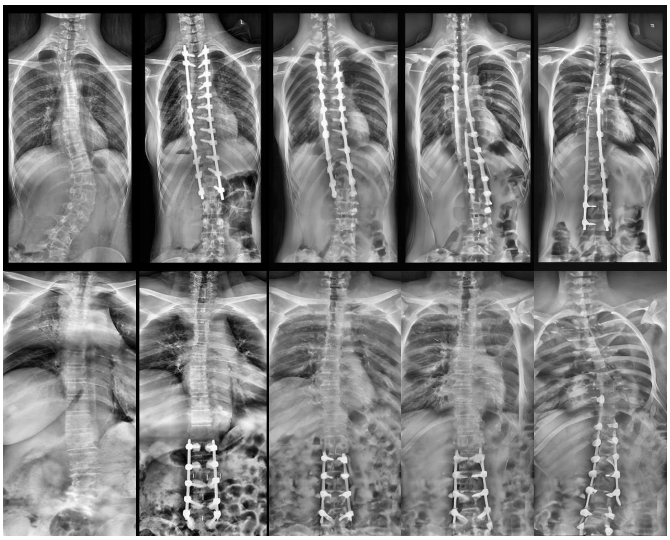
Table I presents the quantitative results of postoperative radiograph synthesis. We compare ScolioLDM with several representative image synthesis methods, including ControlNet [20], BBDM [33], Fast-DDPM [34], ControlNeXt [32],

TABLE I
QUANTITATIVE COMPARISON ON POSTOPERATIVE RADIOGRAPH SYNTHESIS.

Method	SSIM ↑	PSNR ↑	MAE ↓	GradMAE ↓	NCC ↑
ControlNet [20]	0.831 ± 0.009	19.726 ± 1.121	0.036 ± 0.005	0.054 ± 0.005	0.877 ± 0.03
pix2pix [13]	0.846 ± 0.007	19.432 ± 0.861	0.035 ± 0.004	0.061 ± 0.006	0.861 ± 0.03
ControlNeXt [32]	0.841 ± 0.010	19.179 ± 1.140	0.037 ± 0.006	0.059 ± 0.007	0.864 ± 0.03
BBDM [33]	$\mathbf{0.856 \pm 0.010}$	19.654 ± 1.118	0.034 ± 0.005	0.053 ± 0.006	0.869 ± 0.03
Fast-DDPM [34]	0.810 ± 0.011	17.539 ± 1.987	0.049 ± 0.015	0.067 ± 0.012	0.843 ± 0.06
ScolioLDM	0.841 ± 0.011	$\mathbf{20.436 \pm 1.322}$	$\mathbf{0.032 \pm 0.006}$	$\mathbf{0.051 \pm 0.006}$	$\mathbf{0.892 \pm 0.03}$

TABLE II
ABLATION STUDY ON POSTOPERATIVE RADIOGRAPH SYNTHESIS.

Variant	SSIM ↑	PSNR ↑	MAE ↓	GradMAE ↓	NCC ↑
w/o MGCM	0.836 ± 0.009	19.951 ± 1.142	0.034 ± 0.005	0.053 ± 0.005	0.882 ± 0.035
w/o postoperative curve map	0.822 ± 0.008	19.226 ± 1.007	0.039 ± 0.005	0.057 ± 0.006	0.857 ± 0.039
ScolioLDM	$\mathbf{0.841 \pm 0.011}$	$\mathbf{20.436 \pm 1.322}$	$\mathbf{0.032 \pm 0.006}$	$\mathbf{0.051 \pm 0.006}$	$\mathbf{0.892 \pm 0.035}$



(a) preoperative radiograph (b) postoperative radiograph (GT) (c) ScolioLDM (d) ScolioLDM w/o MGCM (e) ScolioLDM w/o curve map

Fig. 4. Visual comparison of postoperative radiograph synthesis across different ablated versions of ScolioLDM. From left to right: (a) real preoperative radiograph, (b) real postoperative radiograph (GT), (c) ScolioLDM (ours), (d) ScolioLDM without MGCM, and (e) ScolioLDM without postoperative curve map.

and pix2pix [13]. Overall, ScolioLDM achieves the best overall performance across most evaluation metrics, demonstrating superior reconstruction quality and anatomical consistency.

Specifically, ScolioLDM achieves the highest PSNR of 20.436 (±1.322) and NCC of 0.892 (±0.03), while obtaining the lowest MAE of 0.032 (±0.006) and GradMAE of 0.051 (±0.006). These results indicate that ScolioLDM provides improved image fidelity, stronger intensity consistency, and better preservation of anatomical structures. Although BBDM [33] achieves the highest SSIM score of 0.856 (±0.010), its PSNR, MAE, GradMAE, and NCC remain inferior to those of ScolioLDM, suggesting that the proposed method achieves a better balance between structural preservation, pixel-level accuracy, and visual realism.

Fig. 5 further presents qualitative comparison results. Compared with competing methods, ScolioLDM generates postoperative radiographs with clearer vertebral structures, more realistic spinal correction morphology, and more anatomically consistent screw–rod implant placement, showing the closest resemblance to the GT postoperative radiographs. In contrast, ControlNet [20] produces blurred vertebral structures and less accurate implant positioning, while pix2pix [13] fails to preserve realistic postoperative spinal alignment and introduces over-smoothed anatomical details. ControlNeXt [32] improves the global postoperative appearance to some extent, but still suffers from unclear vertebral boundaries and locally inconsistent screw–rod placement. BBDM [33] shows residual inconsistencies around the corrected spine and implant regions, whereas Fast-DDPM [34] introduces noticeable artifacts and distorted spinal morphology.

Overall, the quantitative and qualitative results demonstrate that the proposed framework can synthesize postoperative radiographs with high anatomical fidelity, structural consistency, and visual realism.

E. Ablation Study

To evaluate the effectiveness of different components in the proposed framework, we conduct ablation experiments on postoperative radiograph synthesis. Specifically, we compare the complete ScolioLDM with two ablated variants: ScolioLDM without MGCM and ScolioLDM without postoperative curve map guidance.

Table II presents the quantitative ablation results. Overall, the complete model achieves the best performance across all evaluation metrics, demonstrating the effectiveness of both MGCM and postoperative curve map guidance. Specifically, ScolioLDM achieves the highest SSIM of 0.841 (±0.011), PSNR of 20.436 (±1.322), and NCC of 0.892 (±0.035), while obtaining the lowest MAE of 0.032 (±0.006) and GradMAE of 0.051 (±0.006), indicating improved reconstruction quality and anatomical consistency. Removing MGCM leads to decreased performance across all metrics, with SSIM, PSNR, and NCC decreasing to 0.836 (±0.009), 19.951 (±1.142), and 0.882 (±0.035), respectively. Furthermore, removing the

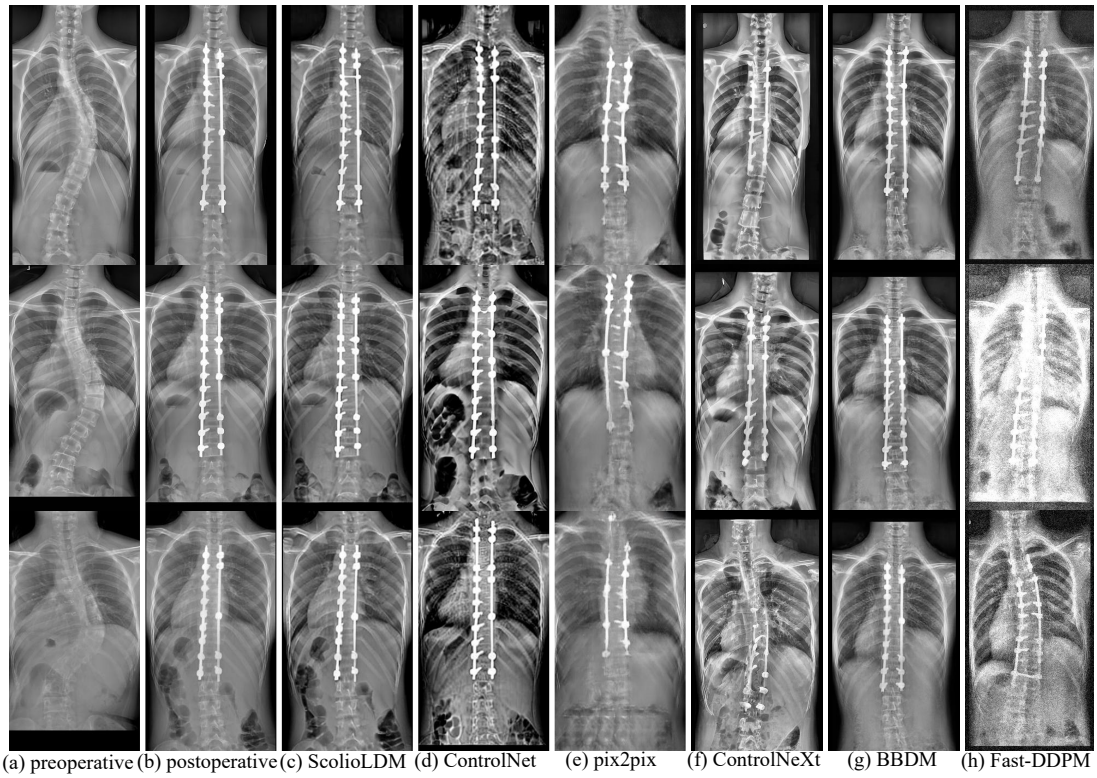


Fig. 5. Visual comparison of postoperative radiograph synthesis generated by different methods. From left to right: (a) real preoperative radiograph, (b) real postoperative radiograph (GT), (c) ScolioLDM, (d) ControlNet [20], (e) pix2pix [13], (f) ControlNeXt [32], (g) BBDM [33], and (h) Fast-DDPM [34]. Compared with competing approaches, ScolioLDM generates postoperative radiographs with more realistic spinal correction, clearer vertebral structures, and more anatomically consistent screw–rod implant placement, showing the closest resemblance to the GT postoperative radiographs in column (b).

postoperative curve map guidance results in the largest performance degradation, with SSIM, PSNR, and NCC decreasing to 0.822 ($\pm$0.008), 19.226 ($\pm$1.007), and 0.857 ($\pm$0.039), respectively, highlighting its critical role in postoperative radiograph synthesis.

Fig. 4 further shows the qualitative comparison results. The complete model generates postoperative radiographs with clearer vertebral structures, more realistic spinal alignment correction, and more anatomically consistent screw–rod implant configurations. In contrast, removing MGCM introduces local inconsistencies in vertebral morphology and implant placement, while the variant without postoperative curve map guidance produces more noticeable deviations in spinal curvature and structural details.

Overall, both quantitative and qualitative results demonstrate that MGCM and postoperative curve map guidance are essential components of the proposed framework, significantly improving the anatomical fidelity and visual realism of synthesized postoperative radiographs.

IV. Discussion

A. Input Design and Ongoing Work for Automatic Curve Map Generation

In this study, the input of ScolioLDM consists of both the preoperative radiograph and a spine curve map constructed from manually annotated vertebral landmarks. While manual landmark annotation ensures high structural accuracy and provides reliable morphological guidance for model training, it requires expert clinical input and introduces additional preprocessing overhead. To further enhance the clinical practicality of the proposed framework, ongoing research is being conducted to develop a fully automatic vertebral landmark detection pipeline, which aims to identify the four corner points of each vertebra directly from raw preoperative radiographs. Based on these detected landmarks, the spinal centerline can be automatically estimated, enabling end-to-end generation of the curve map without manual intervention. This direction will improve the automation and usability of the proposed method, facilitating its integration into routine preoperative assessment workflows.

B. Role and Significance of Curve Map and MGCM Design

The curve map and MGCM form the core morphological guidance mechanism for anatomically consistent postoperative radiograph synthesis. As a simplified structural prior, the curve map explicitly encodes spinal trajectory and curvature, suppressing irrelevant image information while emphasizing clinically relevant geometry. This constraint guides the diffusion model toward plausible spinal deformations and reduces unrealistic artifacts. Building upon this prior, MGCM further enhances morphological modeling through structural refinement, spatial attention modulation, and adaptive deformation

alignment, improving the reconstruction of vertebral alignment and implant configurations. Together, they bridge image appearance and spinal morphology, enabling anatomically plausible and clinically meaningful postoperative radiograph synthesis.

C. Limitations and Future Work

Despite the promising results, several limitations remain. The current curve map generation relies on manual annotation, limiting scalability. In addition, the single-center and relatively small dataset may affect generalizability and increase the risk of overfitting. Furthermore, ScolioLDM currently supports only 2D AP radiograph synthesis and cannot generate lateral views or perform 3D spinal reconstruction. Future work will focus on developing a fully automated landmark detection and curve map generation pipeline, expanding validation to multi-center cohorts with larger sample sizes, and extending ScolioLDM to multi-view synthesis and 3D spinal modeling. Ultimately, we aim to establish a clinically deployable AI system for intuitive, accurate, and automated preoperative planning in AIS patients.

V. Conclusions

In this paper, we propose ScolioLDM, a LDM framework for AIS postoperative radiograph synthesis. Taking a preoperative radiograph as input, our model directly synthesizes realistic postoperative radiographs. To handle complex spinal deformations and implant configurations, we introduce an MGCM and adopt a postoperative curve map as structural guidance, ensuring anatomical consistency and structural accuracy. Extensive experiments demonstrate that ScolioLDM outperforms mainstream methods in both image fidelity and anatomical plausibility. We also construct the first paired pre- and postoperative AIS radiograph dataset to serve as a benchmark. Overall, ScolioLDM provides an intuitive, automated tool to assist clinical spinal surgery planning.

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